Energy and Thermal Evaluation of Sage Thor-Blade Edge Computing Platform for Continuous Foundation Model Inference.

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Introduction/Motivation

- Sage (See Fig. 1) is a nation-scale cyberinfrastructure to enable artificial intelligence at the edge (AI@Edge) for science [1]. Sage edge computing nodes provide in-situ data processing to field-deployed scientific instrument, enabling real-time sensing and data-driven actuation.
- The next generation of Sage edge computing platform, named Sage Thor-Blade, offers powerful GPU computation for deploying foundation models at the edge and AI inference on complex, multidimensional data (e.g., high-resolution images). However, over 100 W power computation of Nvidia Jetson Thor Sage Thor-Blade node will far exceed the current design specifications (See Fig. 2), requiring a smart balance between inference capability and thermal and energy constraints.
- This work presents a thermal characterization of Sage Thor-Blade node under sustained foundational AI inference, with emphasis on the effect of different power modes and cooling configurations. We evaluate AI inference workload and measure the resulting energy consumption and thermal behavior during continuous AI operation. Our analysis compares transient and steady-state thermal responses across workload and cooling conditions, revealing how workload intensity and power model configuration interact with hardware cooling capacity and AI throughput.

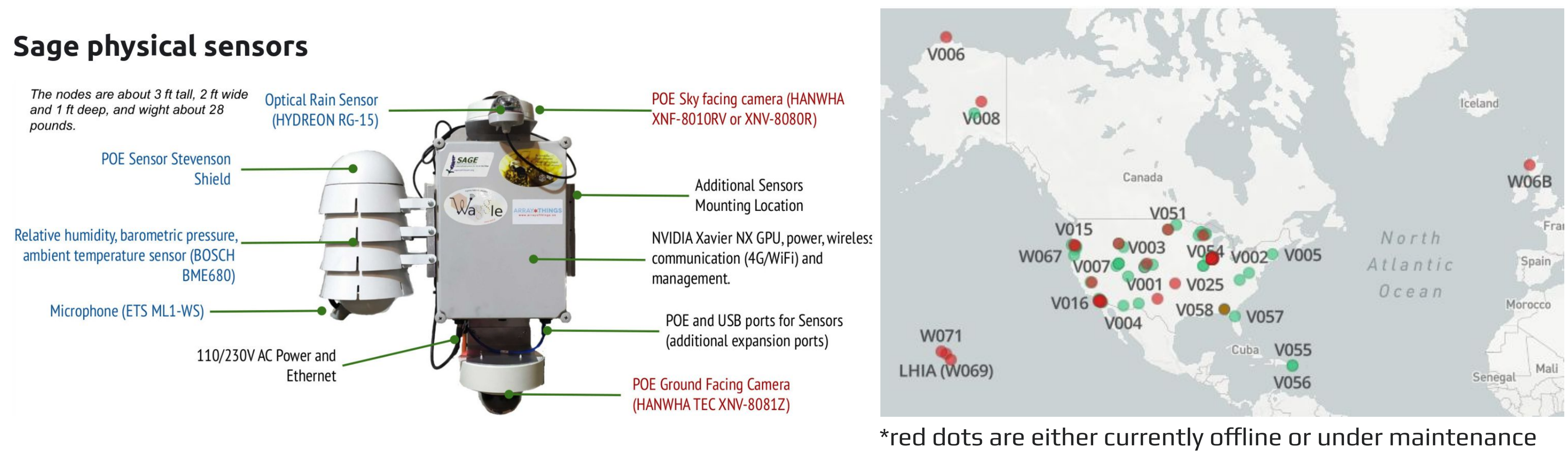


Figure 1. (left) Wild Sage edge computing node and (right) 148 Sage node deployment

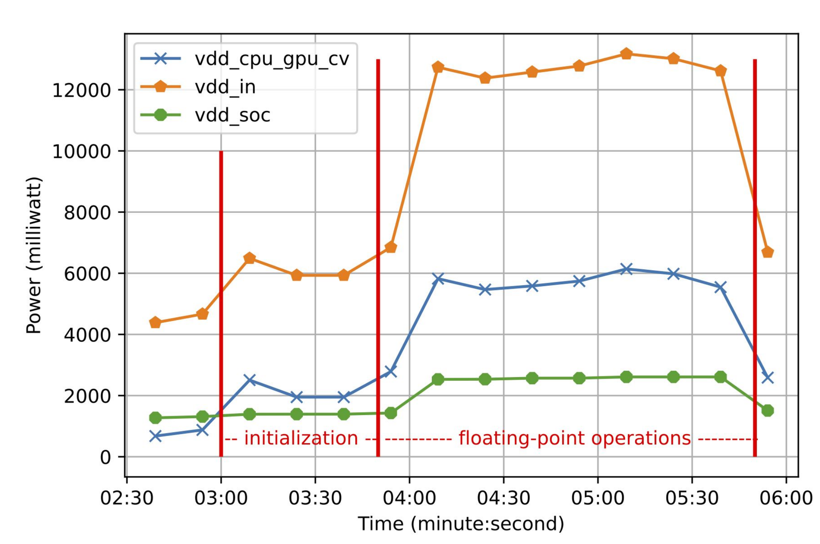


Figure 2. GPU stress test on Nvidia Jetson NX inside the wild Sage node, drawing around 13 W for GPU computation

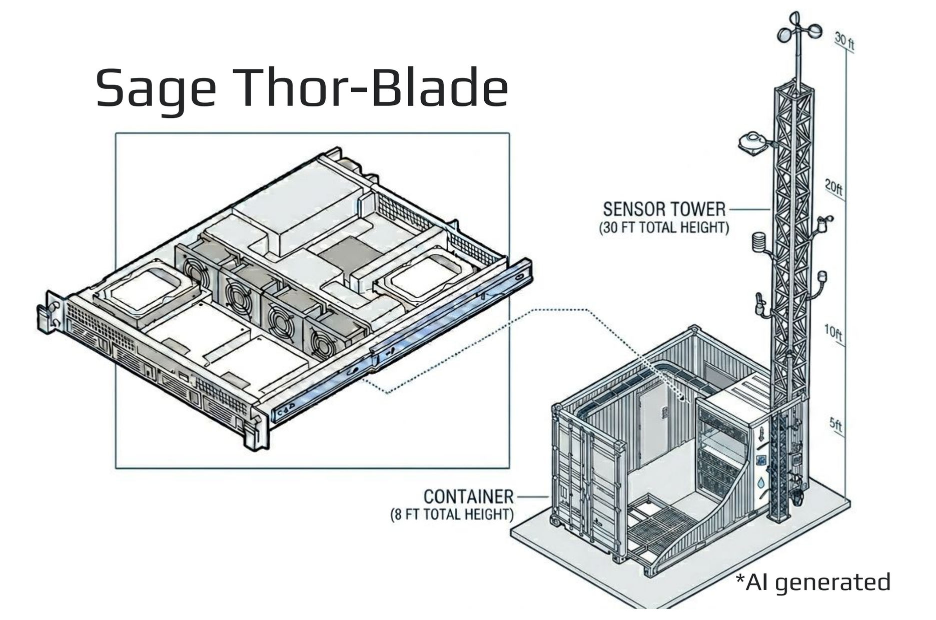


Figure 4. Thor-Blade node enclosed in 2U server chassis gets deployed in an indoor environment for AI computation on live sensor stream

Methodology

- Two Thor-Blade designs: Nvidia Thor Devkit and Thor on AVerMedia carrier board [2], both with active cooling fans (See Fig. 3). The current Thor-Blade is for in-door deployment and equipped with a local network port to connect to networked sensors (See Fig. 4).

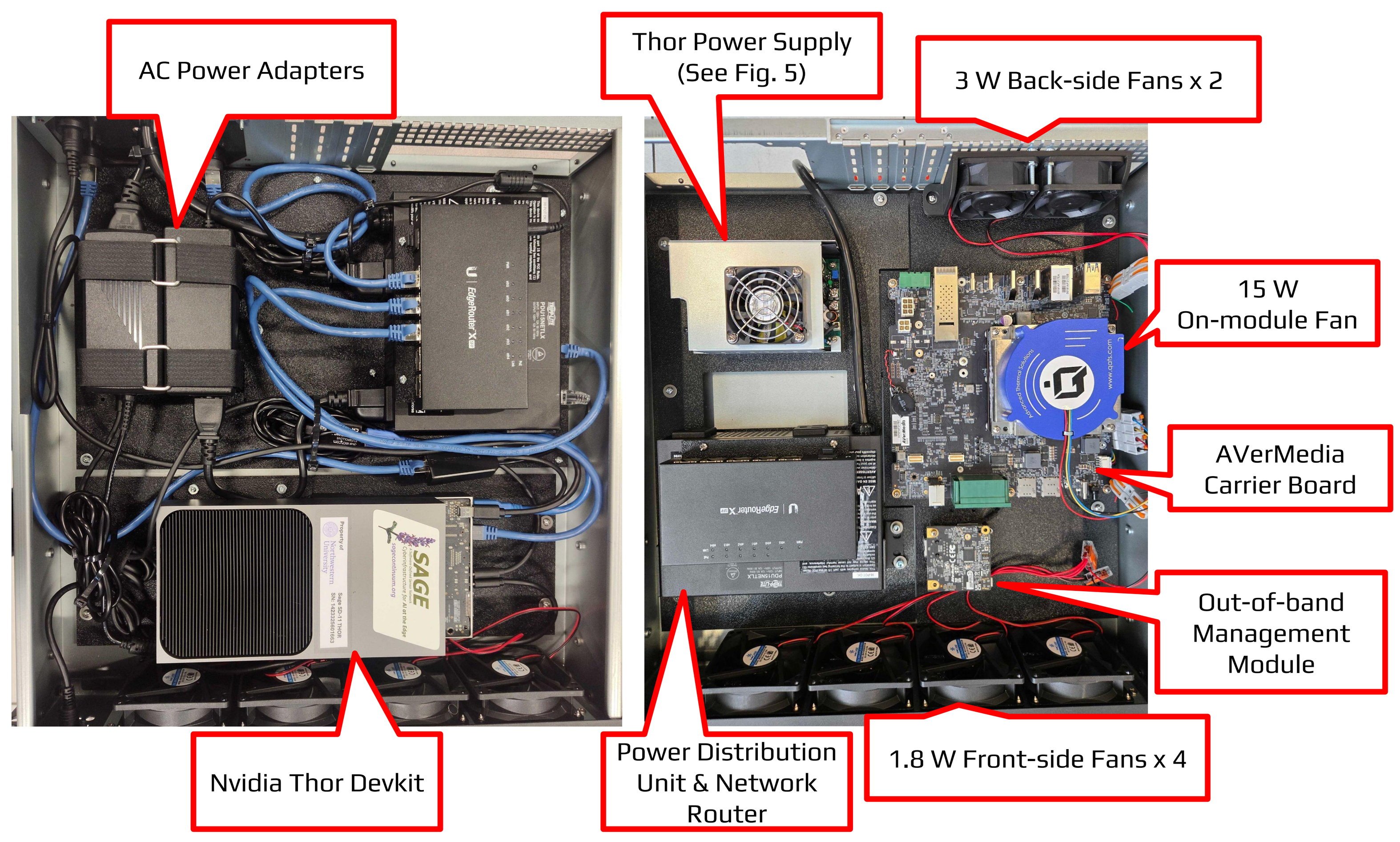


Figure 3. Thor-Blade nodes with (left) Nvidia Devkit and (right) AVerMedia board

- Workload: Continuous reasoning task using gemma3:27b via ollama. Each test runs for 2 hours to observe the steady state of load and node performance.
- Power modes for Thor: MAX, 120W, and 70W [4]. (See Table 1)

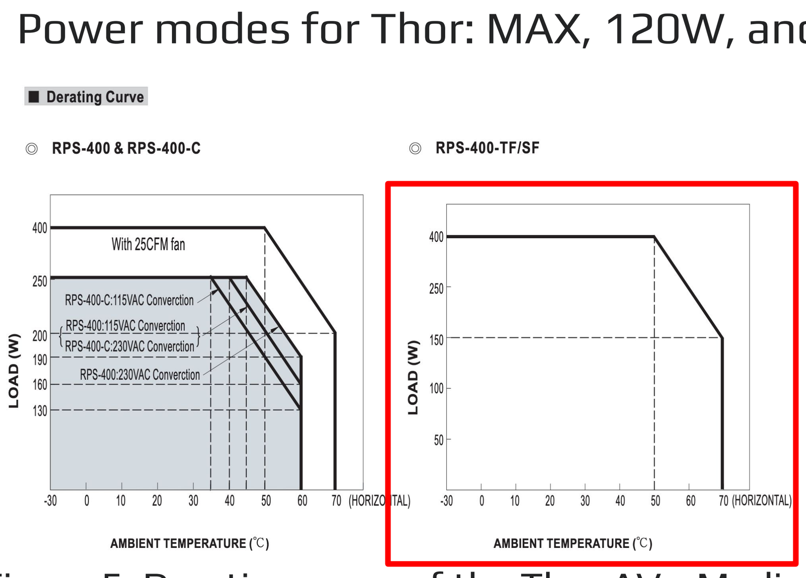
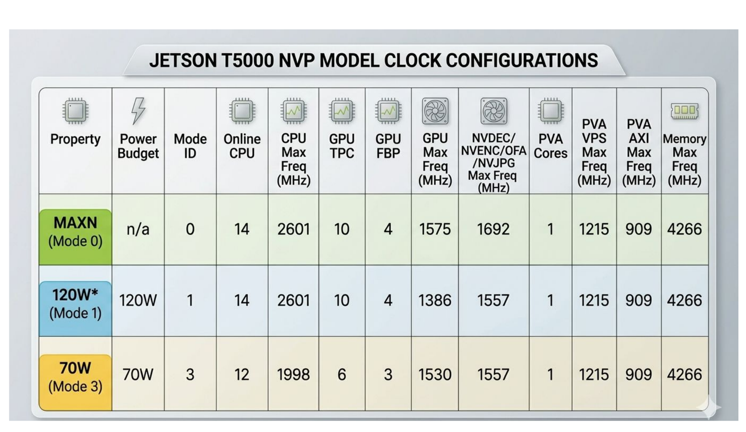


Figure 5. Derating curve of the Thor AVerMedia power supply. The power rating of 150 W at 70 C ensures continuous power supply under heavy AI computation [3].

Table 1. Jetson Thor Power Modes



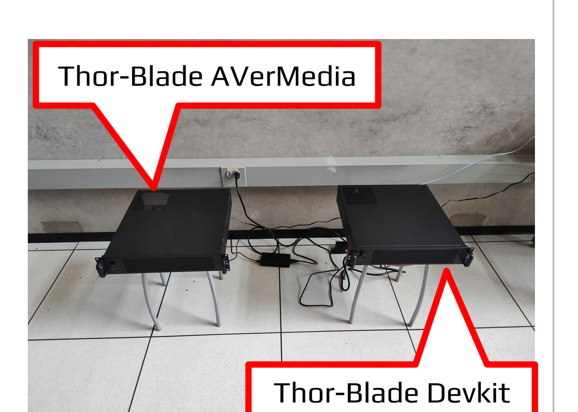
JETSON T5000 NVP MODEL CLOCK CONFIGURATIONS

Property	Power Budget	Mode ID	Online CPU	CPU Max Freq (MHz)	GPU TPC	GPU FBP	GPU Max Freq (MHz)	NVDEC/ NVENC/OFA /NVJPG Max Freq (MHz)	PVA Cores	PVA VPS Max Freq (MHz)	PVA AXI Max Freq (MHz)	Memory Max Freq (MHz)
MAXN (Mode 0)	n/a	0	14	2601	10	4	1575	1692	1	1215	909	4266
120W* (Mode 1)	120W	1	14	2601	10	4	1386	1557	1	1215	909	4266
70W (Mode 3)	70W	3	12	1998	6	3	1530	1557	1	1215	909	4266

*AI generated

Experimental Results

- Sage Thor-Blade Devkit
  - AI throughput: 9.72 - 8.13 tokens/sec
  - Power consumption: 80 - 100 W on average
  - Ambient temperature on SoC: 60 - 70 C
  - Energy efficiency: 0.1 tokens/sec per W on average
- Sage Thor-Blade AVerMedia
  - AI throughput: 9.2 - 10.43 tokens/sec
  - Power consumption: 120 - 140 W on average
  - Ambient temperature on SoC: 58 - 62 C
  - Energy efficiency: 0.078 tokens/sec per W on average



Conclusion

- Foundation AI models bring new capability to edge computing; reasoning on contextual information from sensor stream for high-level scientific analysis.
- Sage next-generation edge computing platform provides sustainable computing environment to foundation AIs for continuous inference at the edge.
- Active cooling mechanism allows the Thor-Blade node to be under thermal-comfort zone and slightly increases AI throughput by sacrificing energy for cooling.

Reference

[1] Sage cyberinfrastructure, https://sagecontinuum.org/
[2] AVerMedia Thor Carrier board, https://professional.avermedia.com/product-detail/D331
[3] Power supply unit for Thor AVerMedia carrier board, https://www.meanwellusa.com/upload/pdf/RPS-400/RPS-400-spec.pdf
[4] Nvidia Jetson Thor Power Modes, https://docs.nvidia.com/jetson/archives/r38.4/DeveloperGuide/SD/PlatformPowerAndPerformance/JetsonThor.html#supported-modes-and-power-efficiency

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